

Representing Functions and Fields

Finitely many numbers for a function, and what each choice loses

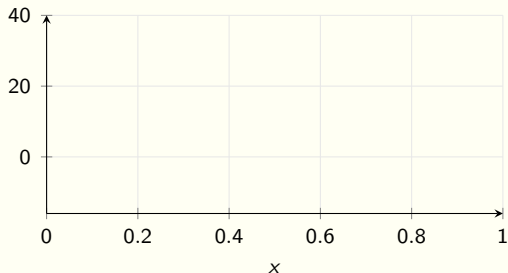
Krishna Kumar

University of Texas at Austin

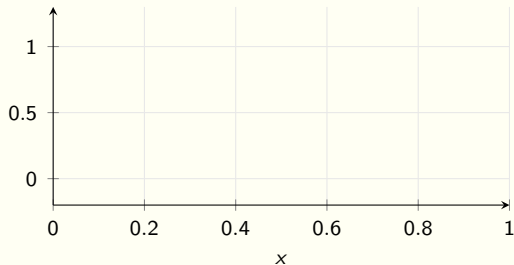
krishnak@utexas.edu

A taut string under a load, $\mathcal{L}u = f$

the load $f(x) = -u''(x)$

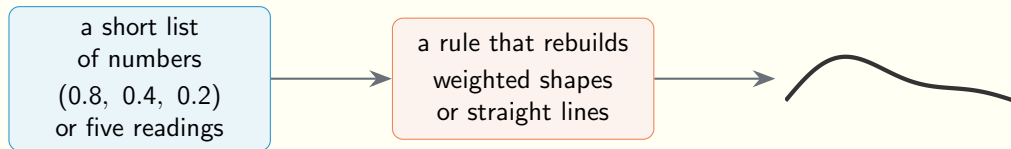


the displacement $u(x)$, pinned at both ends



$$-\frac{d^2 u}{dx^2} = f(x), \quad u(0) = u(1) = 0, \quad \mathcal{L} = -\frac{d^2}{dx^2}$$

A finite list of numbers and a rule that rebuilds the function

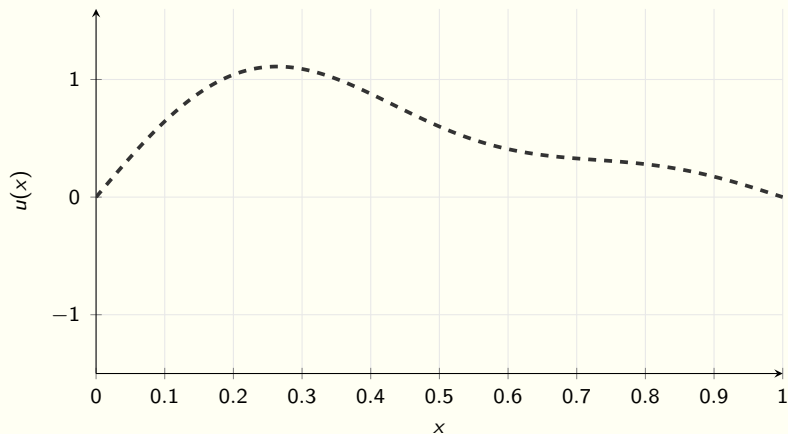


How well can finitely many numbers stand in for a function, and what decides the accuracy?

Outline

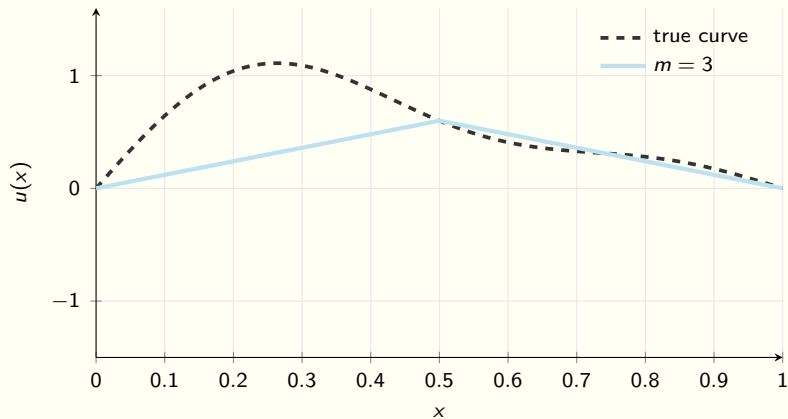
- 1 Pointwise sampling
- 2 Functions as vectors
- 3 Norms and the meaning of error
- 4 Function spaces, L^2 , H^1 , and Hilbert spaces
- 5 Inner products, projection, and orthogonality
- 6 Changing the basis
- 7 Convergence, density, and completeness
- 8 Kernels and point evaluation
- 9 Noisy observations, ridge regression, and Gaussian conditioning
- 10 Approximating a family of functions
- 11 Adaptive representations
- 12 Choosing a representation for the cooling fin
- 13 From functions to fields

Sampling the displacement at fixed stations



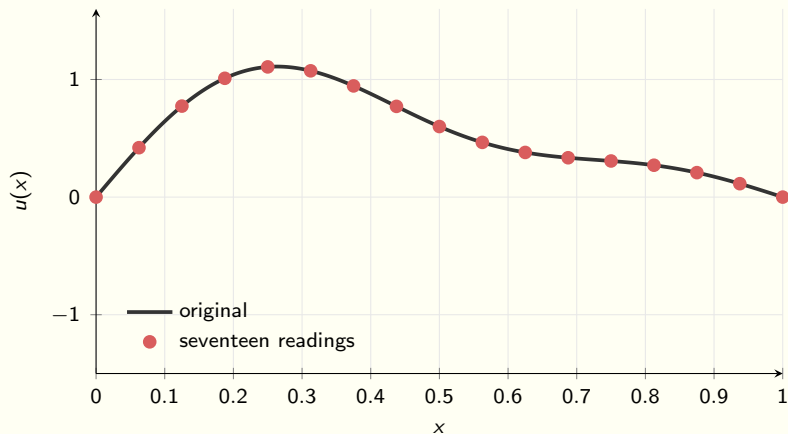
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Convergence of piecewise-linear interpolation with the spacing



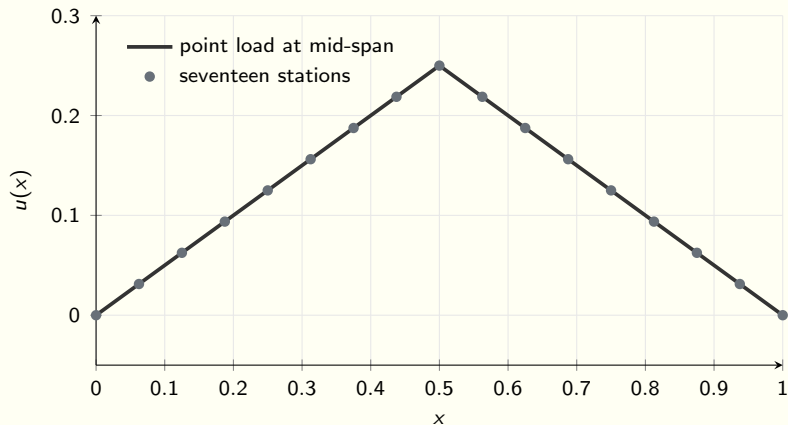
$m = 3, h = 1/2$
73.5 percent

Aliasing of a wave that vanishes at every station

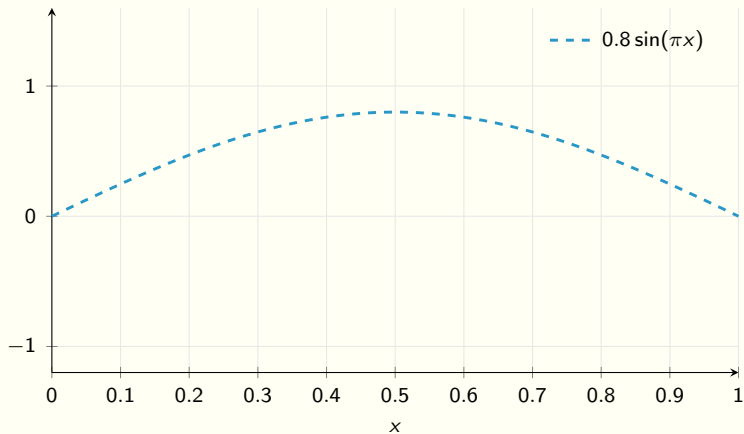


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Grid dependence of sampled representations

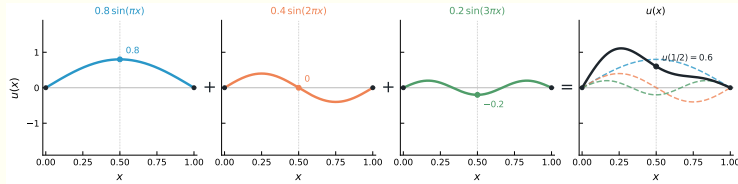


Representing a function with coefficients on sine modes



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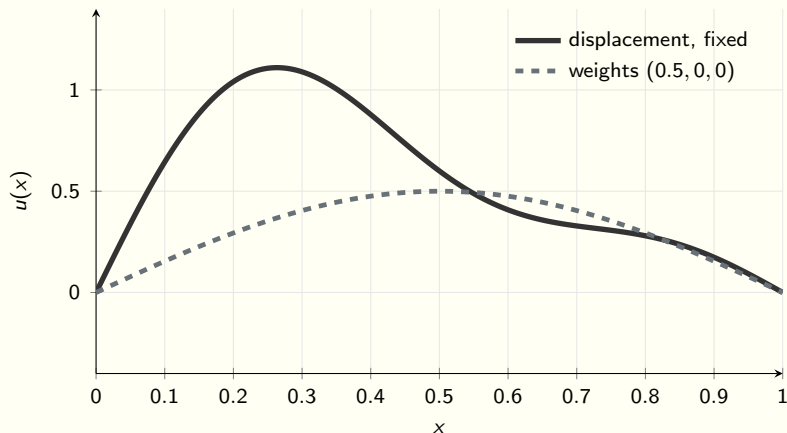
A three-parameter family of string displacements



the family

$$u = a_1 \sin(\pi x) + a_2 \sin(2\pi x) + a_3 \sin(3\pi x)$$

Matching the displacement with three weights

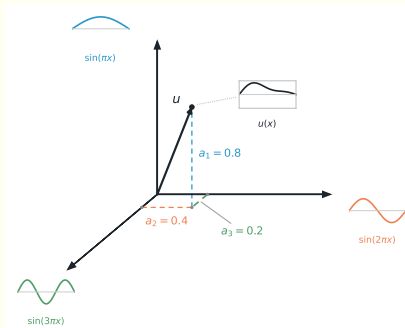
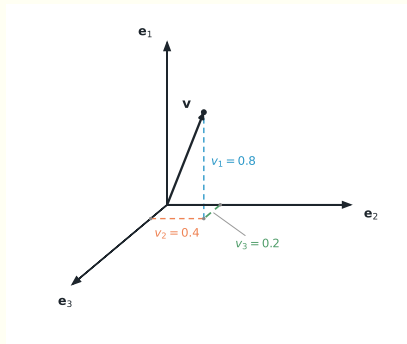


$\sup, L^2, H^1$ errors
68.5, 58.8, 81.5
percent

L^2 distance
0.381

[Try it live](#)

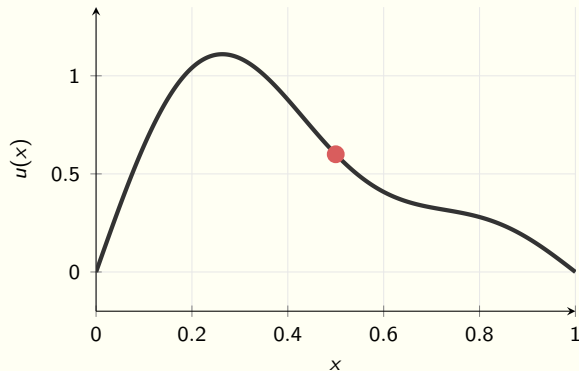
The coefficient list as coordinates in a vector space



$$u + v = \sum_{i=1}^3 (a_i + b_i) \phi_i,$$

$$\alpha u = \sum_{i=1}^3 (\alpha a_i) \phi_i$$

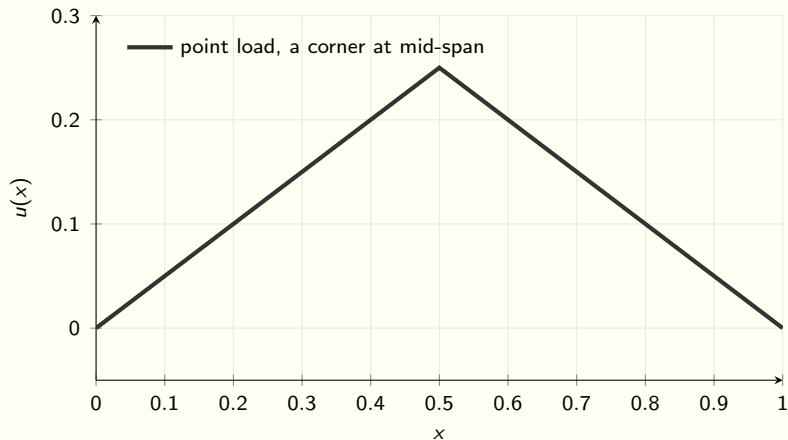
Recovering the coefficients from three readings



$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} & 1 & 1/\sqrt{2} \\ 1 & 0 & -1 \\ 1/\sqrt{2} & -1 & 1/\sqrt{2} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$$

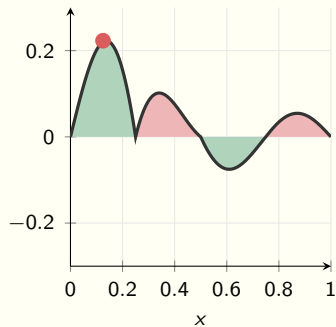
mid-span alone, $y_2 = u(\frac{1}{2})$
 $a_1 - a_3 = 0.6$, **nothing**
about a_2

Approximating a function outside the span



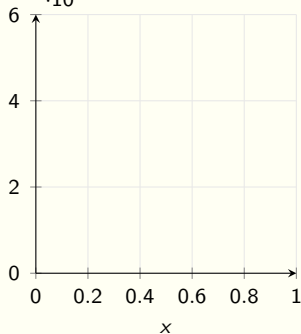
Measuring error in the sup, L^2 , and H^1 norms

error $e = u - v$



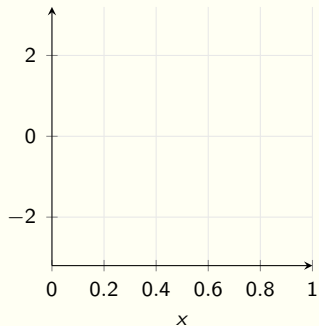
largest gap, sup norm
0.22

squared error e^2
 $\cdot 10^{-2}$



root of the area, L^2 norm
0.093

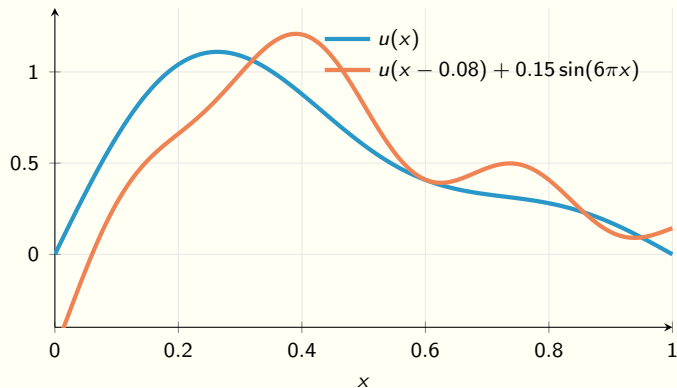
slope error e'



root-mean-square slope,
 H^1 seminorm
1.20

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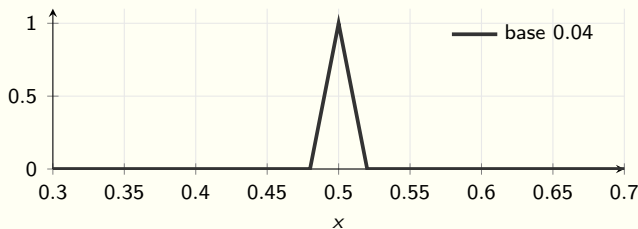
The L^2 distance



$$\|u - g\|_{L^2} = \left(\int_0^1 (u - g)^2 dx \right)^{1/2}$$

[norms notebook](#)

Disagreement between the sup, L^2 , and H^1 norms



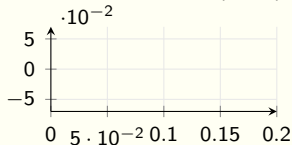
spike, base 0.04

sup 1, L^2 0.115,

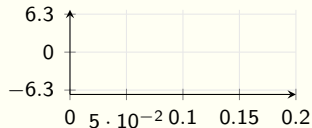
energy 50

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the wiggle $0.05 \sin(40\pi x)$



its slope $2\pi \cos(40\pi x)$



Definition of a norm and a seminorm

Norm

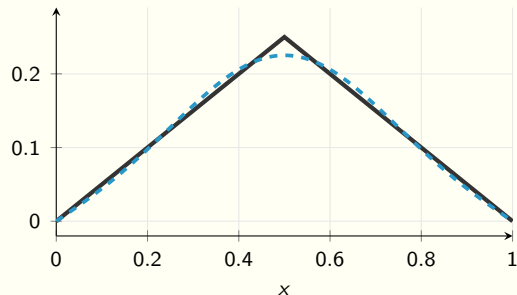
$$\|u\| = 0 \text{ only for } u = 0, \quad \|\alpha u\| = |\alpha| \|u\|, \quad \|u + v\| \leq \|u\| + \|v\|.$$

Seminorm

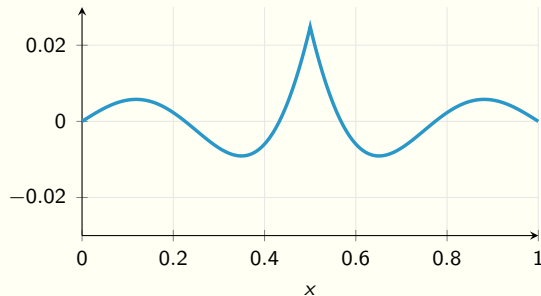
$$|c|_{H^1} = \left(\int_0^1 (c')^2 dx \right)^{1/2} = 0 \quad \text{for every constant } c, \quad \text{and on the pinned string the only constant is } c = 0.$$

Closest three-sine sums to the corner in the L^2 and sup norms

the corner (black), closest in L^2 (blue) and in sup (orange)

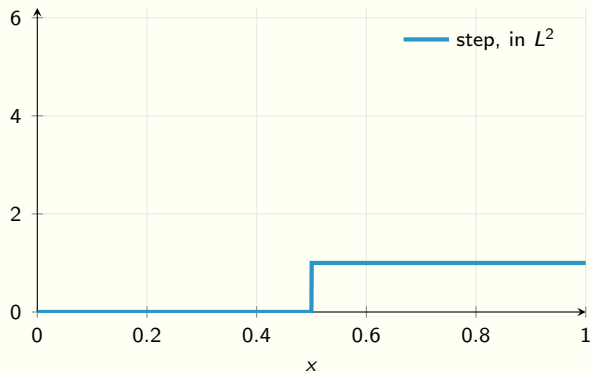


the gap $u - v$



L^2 -closest,
(0.2026, 0, -0.0225)
**sup 9.9, L^2 4.8, H^1
31.5 percent**

The space L^2 of square-integrable functions



$$\|f\|_{L^2}^2 = \int_0^1 f(x)^2 dx < \infty$$

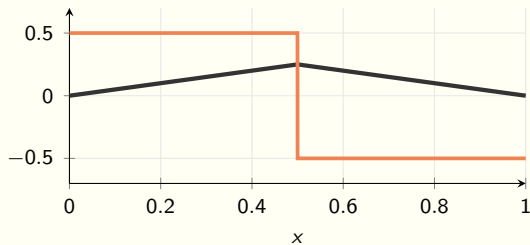
$$\langle f, g \rangle = \int_0^1 fg dx$$

two functions equal except
at one point

L^2 distance 0

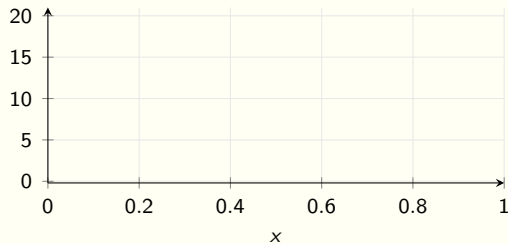
The space H^1 of finite-energy functions

the corner $\min(x, 1-x)/2$ and its slope, in H_0^1

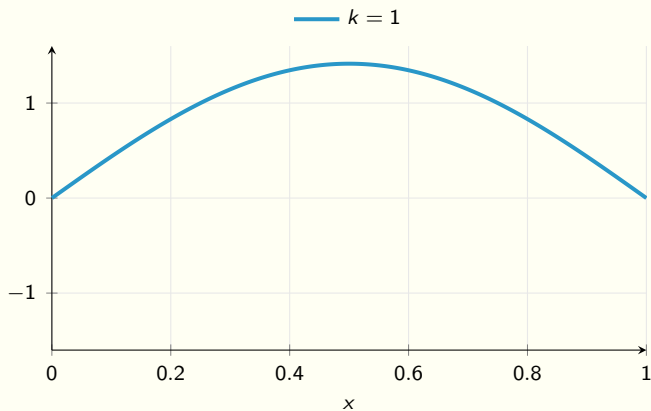


$$\|u\|_{H^1}^2 = \int_0^1 u^2 dx + \int_0^1 (u')^2 dx, \quad \langle u, v \rangle_{H^1} = \int_0^1 (uv + u'v') dx$$

ramps toward a step, and their slopes $1/w$



Equal L^2 size, growing H^1 size



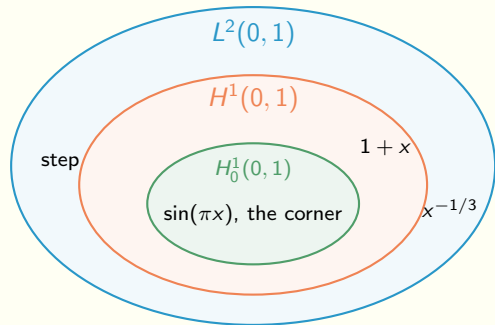
$$\phi_k(x) = \sqrt{2} \sin(k\pi x)$$

$$\|\phi_k\|_{L^2} = 1, \quad |\phi_k|_{H^1} = k\pi$$

$$|\phi_k|_{H^1} \text{ at } k = 1$$

$$3.14$$

Hilbert spaces and the nesting $H_0^1 \subset H^1 \subset L^2$



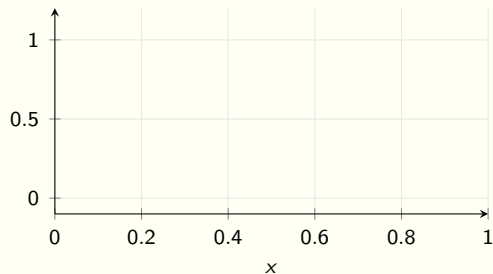
Hilbert space

A vector space with an inner product in which every Cauchy sequence converges to a member of the space.

Hilbert spaces

$\mathbb{R}^n$, $L^2(0,1)$, $H^1(0,1)$,
 $H_0^1(0,1)$

The space H_0^1 and the Poincaré inequality



$$H_0^1(0, 1) = \{ u \in H^1(0, 1) : u(0) = u(1) = 0 \}$$

$$\|u\|_{L^2} \leq \frac{1}{\pi} |u|_{H^1} \quad \text{for } u \in H_0^1(0, 1)$$

The spaces L^2 , H^1 , and H_0^1 compared

space	$u \in L^2$	$\nabla u \in L^2$	inner product	boundary values	role in $-\Delta u = f$
$L^2(\Omega)$					
$H^1(\Omega)$					
$H_0^1(\Omega)$					

Compactness from bounded energy

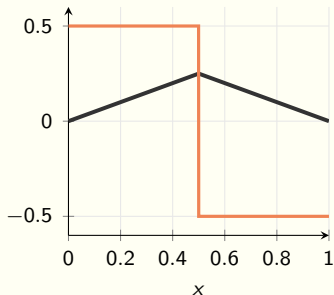
$$u = \sum_{k \geq 1} c_k \sqrt{2} \sin(k\pi x), \quad |u|_{H^1}^2 = \sum_{k \geq 1} (k\pi)^2 c_k^2$$

unit ball of L^2

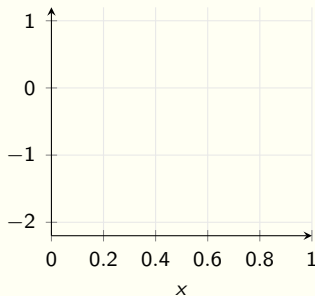
**the modes stay $\sqrt{2}$
apart, no
subsequence
converges**

Regularity from the load to the displacement

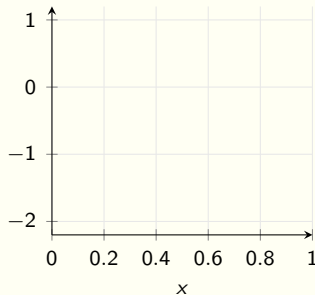
point load, $u \in H_0^1$



step load, $u \in H^2$



uniform load, $u = x(1 - x)$



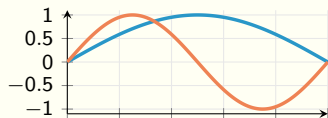
u black, u' orange, u'' green

point load
slope jumps,
 $c_k \sim 1/k^2$

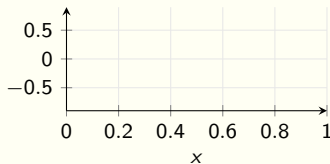
The inner product as signed overlap

$f = \sin(\pi x)$ blue, g orange, and below them fg , green where positive and red where negative

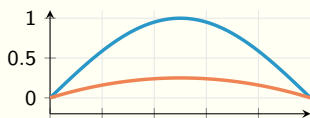
$g = \sin(2\pi x)$



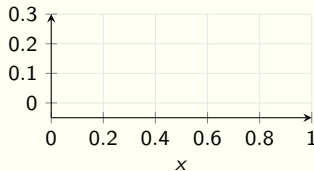
fg , net area 0



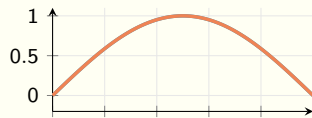
$g = x(1-x)$



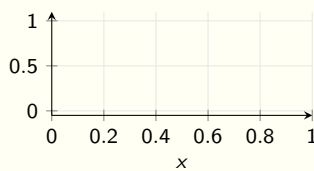
fg , net area 0.129



$g = f$

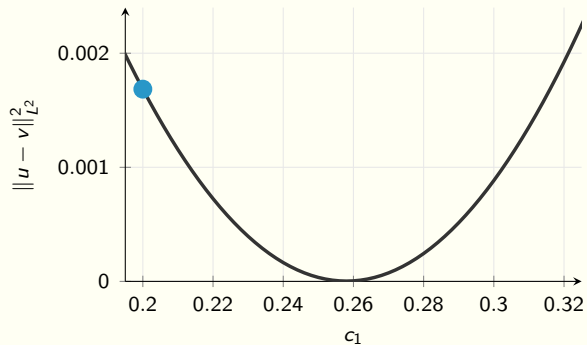
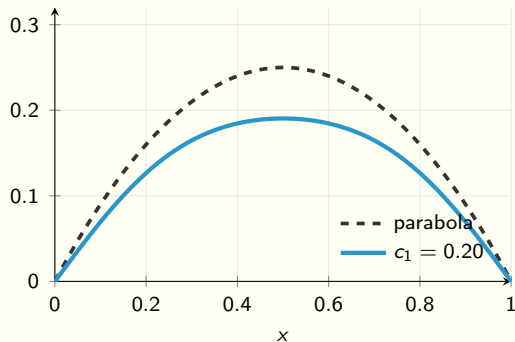


fg , net area 0.500



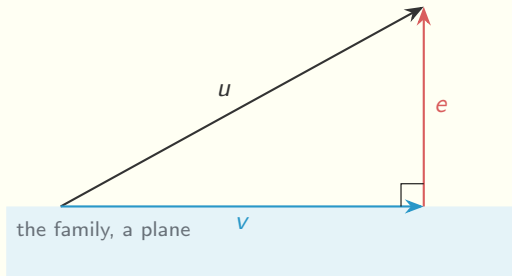
$$\langle f, g \rangle = \int_0^1 f(x) g(x) dx \approx \frac{1}{16} \sum_{i=1}^{16} f(x_i) g(x_i)$$

Minimizing squared error via orthogonal projection



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Uniqueness of the projection and the Pythagorean identity



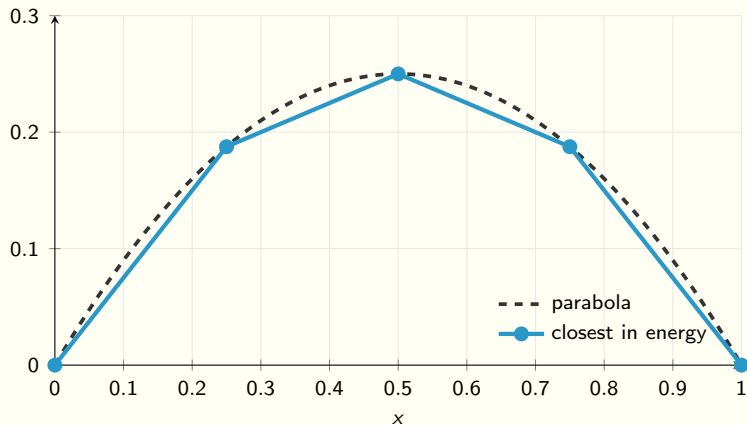
The projection theorem

Projection theorem

In a space with an inner product, every finite-dimensional linear subspace contains one closest member to each target, and that member is the one whose error is orthogonal to the subspace.

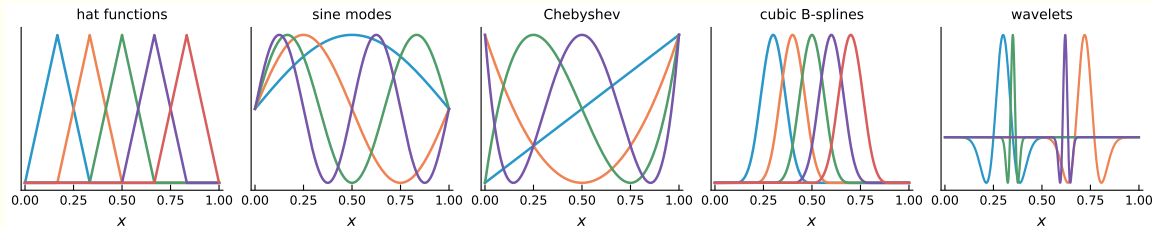
$$\langle u - v, \phi_j \rangle = 0 \quad \text{for every shape } \phi_j, \quad c_k = \frac{\langle u, \phi_k \rangle}{\langle \phi_k, \phi_k \rangle} \text{ when the shapes are orthogonal.}$$

Projection in the L^2 and energy inner products



closest in energy, heights
0.1875, 0.25, 0.1875

Five fixed families of shapes



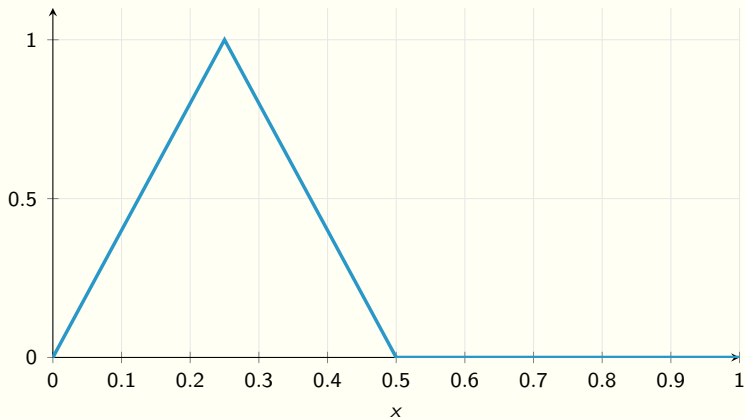
local
hats, one per node

global
**sine modes,
Chebyshev
polynomials**

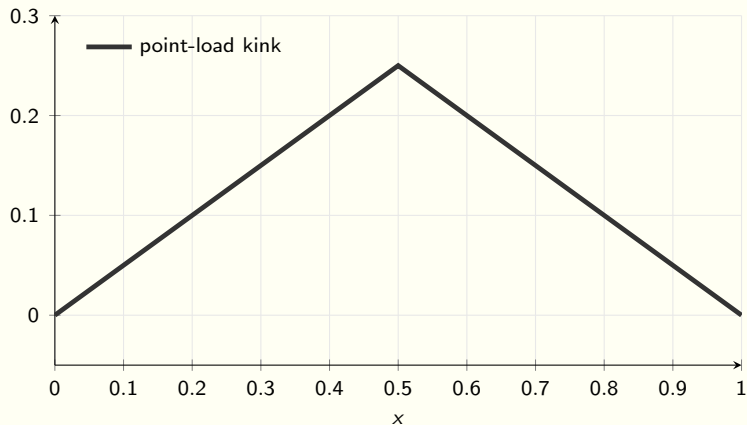
smooth and local
cubic B-splines

local in position and scale
wavelets

Piecewise-linear interpolation as a hat-function expansion

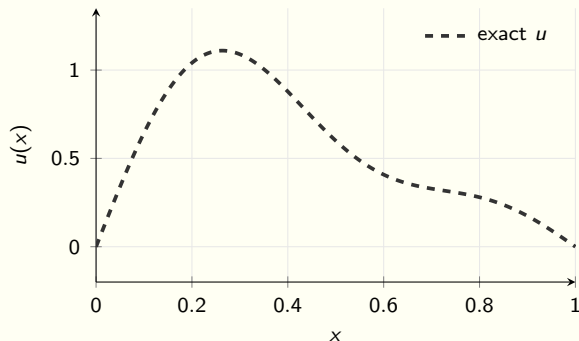


Representing a kink in the hat basis and the sine basis



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Galerkin's method with hat functions

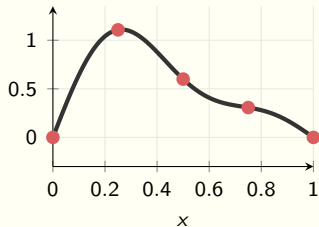


$$\int_0^1 u'_h \phi'_i dx = \int_0^1 f \phi_i dx \quad \text{for every hat } \phi_i$$

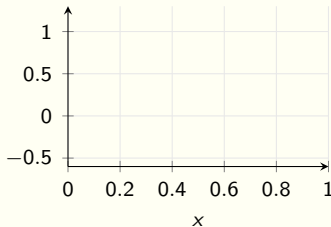
$$\mathbf{K}\mathbf{u} = \mathbf{F}, \quad K_{ii} = \frac{2}{h}, \quad K_{i,i\pm 1} = -\frac{1}{h}$$

Approximating any shape with hat functions

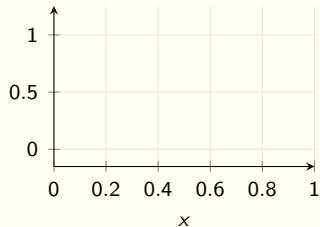
the string, five stations



a bump, seventeen stations



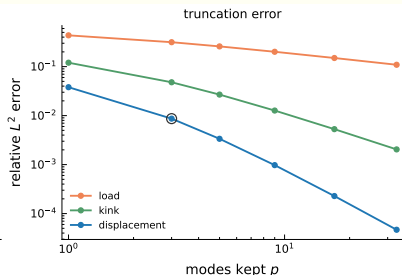
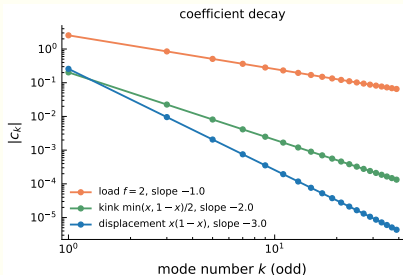
a square pulse, seventeen stations



$$u_h(x) = \sum_i u(x_i) \phi_i(x), \quad \phi_i(x_j) = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

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Decay rates of sine coefficients

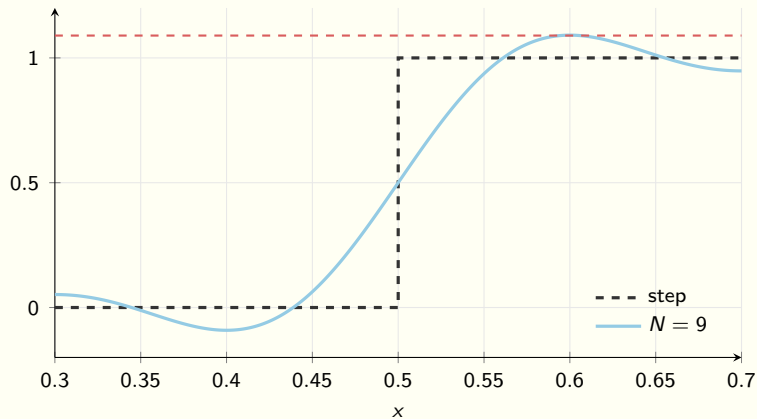


load $f=2$, odd k
 $|c_k| = 8/(k\pi)$, slope -1.0

kink, odd k
 $|c_k| = 2/(k\pi)^2$, slope -2.0

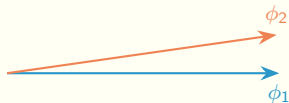
parabola, odd k
 $|c_k| = 8/(k\pi)^3$, slope -3.0

The Gibbs phenomenon at a jump



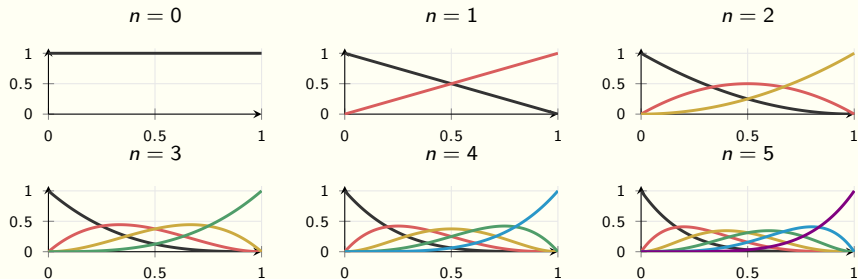
overshoot, $N = 9$
0.091

Conditioning of the Gram matrix for nearly parallel shapes



$$\mathbf{G} = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}, \quad \rho = 0.99, \quad \text{cond } \mathbf{G} = \frac{1 + \rho}{1 - \rho} = 199$$

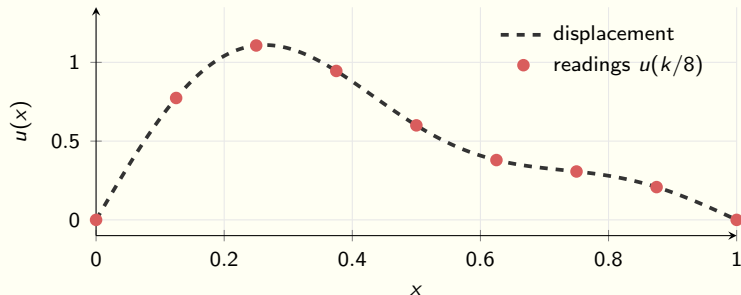
Bernstein basis polynomials of degree 0 to 5



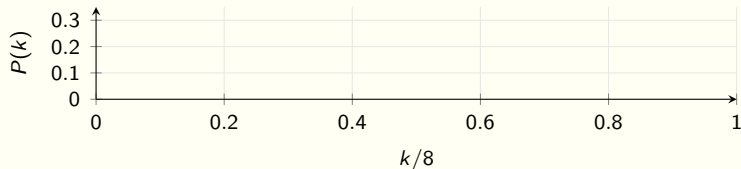
$$B_{i,n}(t) = \binom{n}{i} t^i (1-t)^{n-i}, \quad \sum_{i=0}^n B_{i,n}(t) = 1, \quad B_{i,n}(t) \geq 0$$

$B_{i,n}(t)$ is the probability of exactly i heads in n flips of a coin that lands heads with probability t

Bernstein polynomials for the string displacement



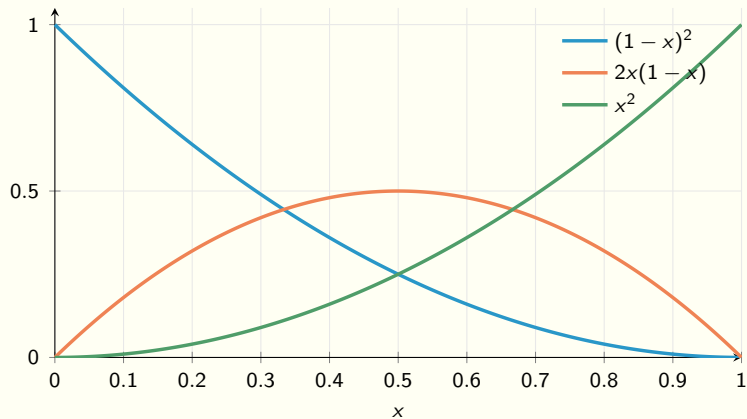
probability of k heads in 8 coins, $x = 0.25$



$$B_n u(x) = \sum_{k=0}^n u\left(\frac{k}{n}\right) b_{k,n}(x)$$

$$b_{k,n}(x) = \binom{n}{k} x^k (1-x)^{n-k}$$

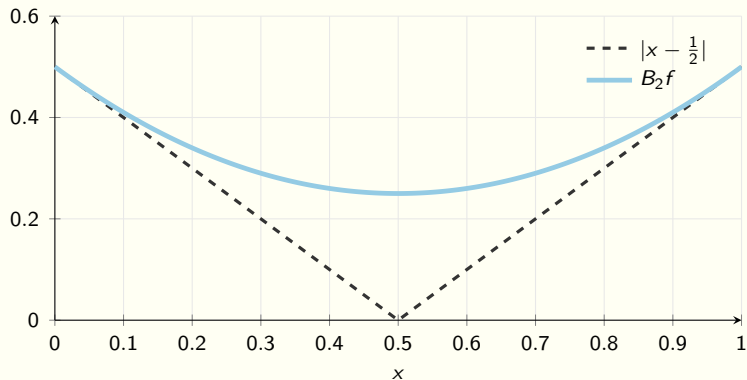
Approximating a corner with Bernstein polynomials



the three hills at mid-span

$\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$

Convergence rate of Bernstein polynomials



gap at the corner, $n = 2$
0.250

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$$B_n f(x) = \sum_{k=0}^n f\left(\frac{k}{n}\right) \binom{n}{k} x^k (1-x)^{n-k}$$

The Weierstrass approximation theorem

Theorem (Weierstrass, 1885)

For every continuous f on $[a, b]$ and every $\epsilon > 0$ there is a polynomial p with $|f(x) - p(x)| < \epsilon$ for every x in $[a, b]$.

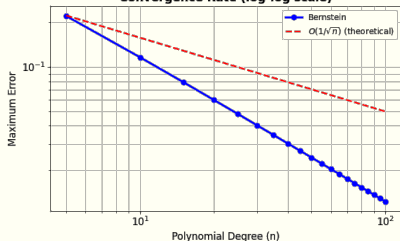
what the theorem does not
say

**the degree, the
weights, or the
slopes**

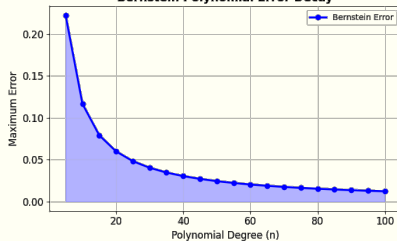
Convergence of Bernstein polynomials for a smooth function

Bernstein Polynomial Convergence: Proof of Weierstrass Theorem

Convergence Rate (log-log scale)



Bernstein Polynomial Error Decay



target
 $\sin(\pi x)$

largest gap, $n = 5$ and
 $n = 100$

0.22 and 0.013

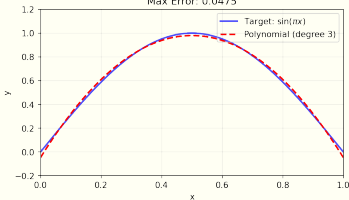
Voronovskaya at mid-span,
 $\pi^2/(8n)$

0.25 and 0.012

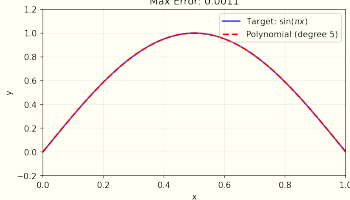
Least-squares polynomial approximation of $\sin(\pi x)$

Polynomial Approximation of $\sin(\pi x)$ (Weierstrass)

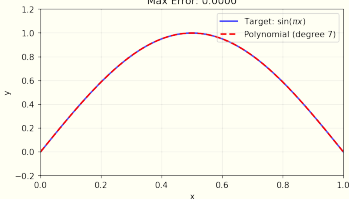
Degree 3 Approximation
Max Error: 0.0475



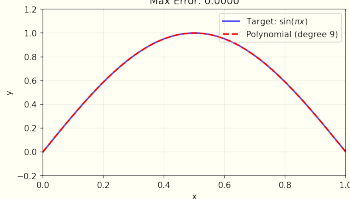
Degree 5 Approximation
Max Error: 0.0011



Degree 7 Approximation
Max Error: 0.0000



Degree 9 Approximation
Max Error: 0.0000



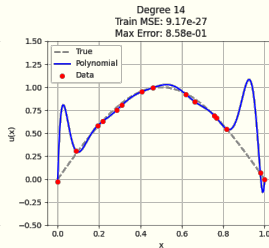
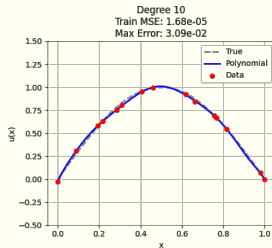
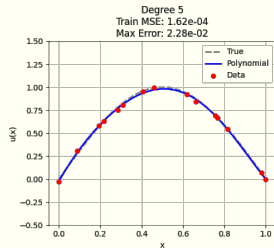
degree 3, largest gap
0.0475

degree 5, largest gap
0.0011

degrees 7 and 9, largest gap
below 0.00005

High-degree interpolation of noisy data

Runge's Phenomenon: Higher Degree $\rightarrow$ Worse Oscillations



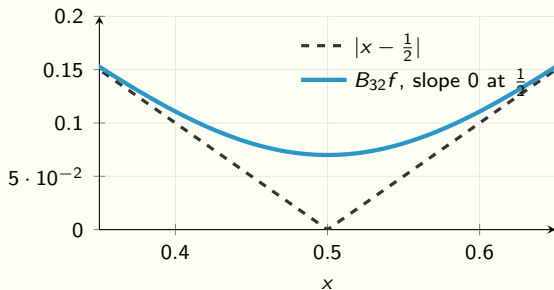
degree 5, least squares
largest gap 0.023

degree 10, least squares
largest gap 0.031

degree 14, through all
fifteen noisy readings
**training mean
squared error**
 9×10^{-27} , **largest
gap 0.86**

What the Weierstrass theorem does not say

$B_{32}f$ beside the corner



Bernstein's error
falls like $1/\sqrt{n}$

best degree- n polynomial
falls like $1/n$

the theorem names
**no polynomial, no
degree, no slope**

Proving density with detectors

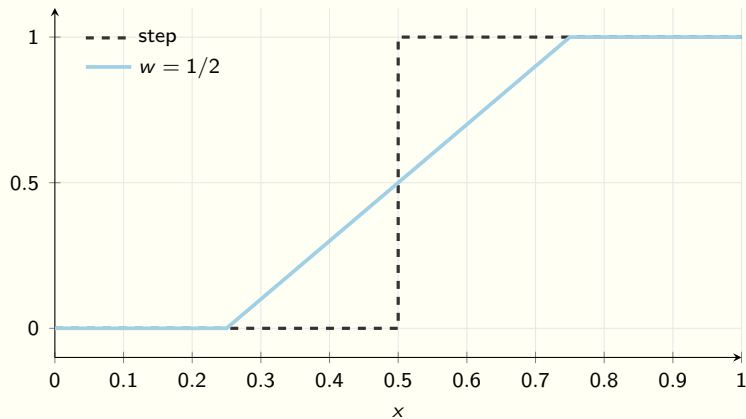
moments forced to zero	response to $\sin(3\pi x)$
0	0.635
2	0.565
4	0.298
8	1.2×10^{-3}
16	3.0×10^{-9}
32	6.9×10^{-16}

$$\int_0^1 x^k d\mu(x) = 0 \quad \text{for every } k$$
$$\implies \int_0^1 f d\mu = 0 \quad \text{for every continuous } f$$

the density proof

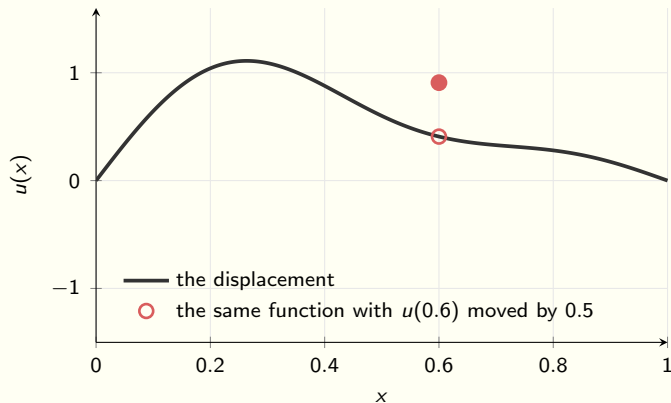
**a detector blind to
every polynomial is
blind to every
continuous function**

Cauchy sequences, completeness, and Hilbert spaces



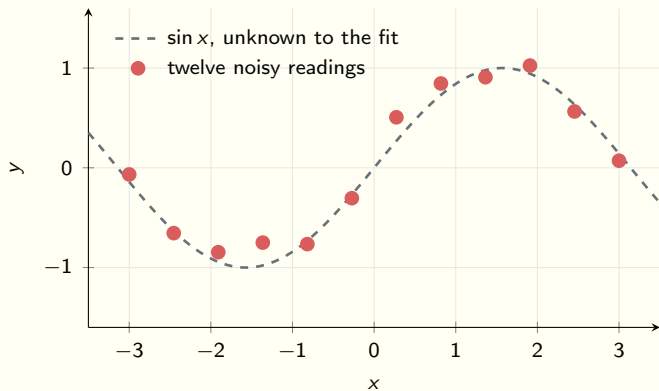
L^2 distance to the step,
 $\sqrt{w/12}$
0.204 at $w = 1/2$

Point evaluation in L^2 and H^1



L^2 distance between the two
0

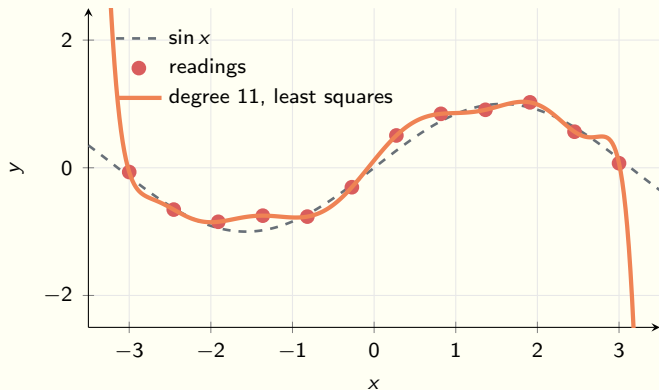
Linear regression by least squares



$$\hat{y}(x) = \sum_j w_j \phi_j(x), \quad \min_{\mathbf{w}} \|\Phi \mathbf{w} - \mathbf{y}\|^2$$

$$\Phi^T \Phi \mathbf{w} = \Phi^T \mathbf{y}$$

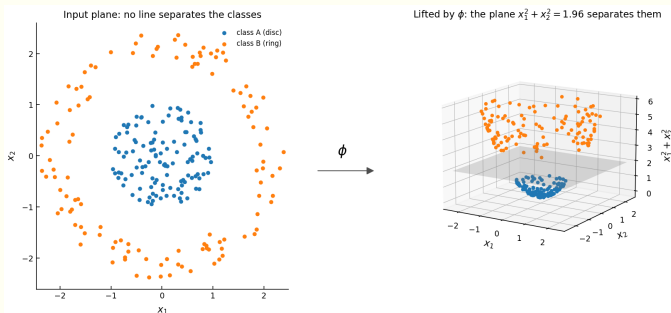
Ridge regression on many features



$$\min_{\mathbf{w}} \|\Phi \mathbf{w} - \mathbf{y}\|^2 + \lambda \|\mathbf{w}\|^2$$
$$(\Phi^T \Phi + \lambda \mathbf{I}) \mathbf{w} = \Phi^T \mathbf{y}$$

degree 11, least squares
error at the data 0,
largest weight 284,
swings to 29.5

A kernel is an inner product of features



$$\phi(x_1, x_2) = (x_1, x_2, x_1^2 + x_2^2)$$

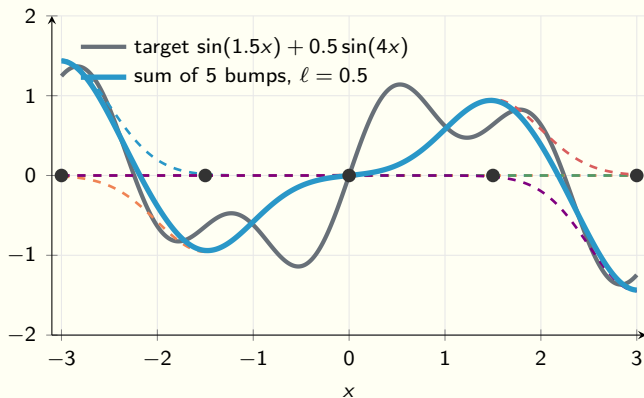
$$k(\mathbf{x}, \mathbf{x}') = \phi(\mathbf{x}) \cdot \phi(\mathbf{x}') = \mathbf{x} \cdot \mathbf{x}' + |\mathbf{x}|^2 |\mathbf{x}'|^2$$

a plane in feature space

$x_3 = 1.96$ separates
disc from ring, no
errors

$\mathbf{x} = (1, 0)$, $\mathbf{x}' = (0.5, 0.5)$
both routes give
1.000

Regression on Gaussian bump features

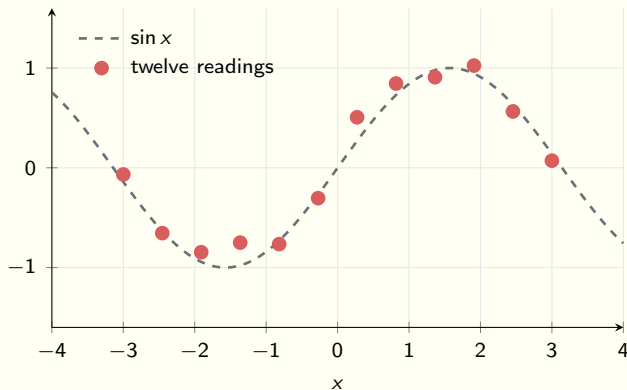


$$\hat{f}(x) = \sum_{j=1}^m w_j e^{-(x-c_j)^2/2\ell^2}$$

5 bumps, $\ell = 0.5$

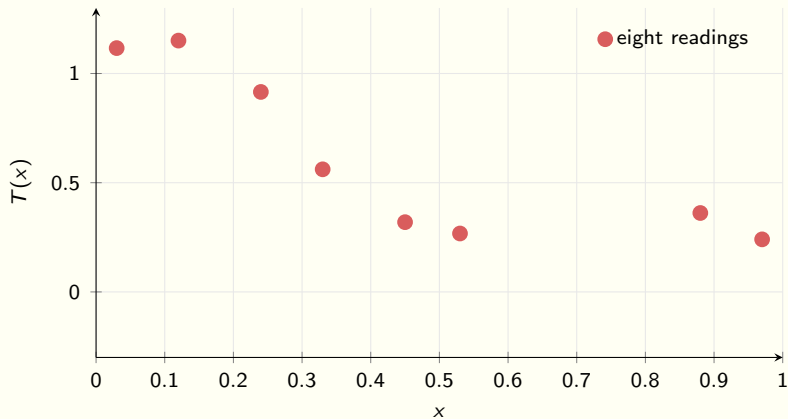
RMSE 0.442

Kernel ridge regression with copies at the data

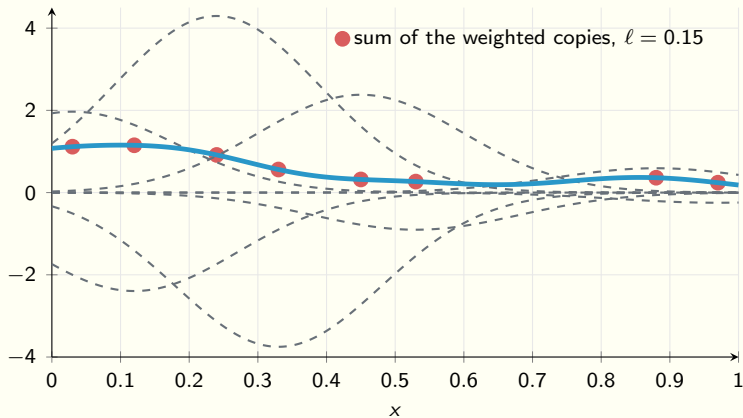


$$\hat{f}(x) = \sum_{i=1}^{12} \alpha_i k(x, x_i), \quad (\mathbf{K} + \lambda \mathbf{I}) \boldsymbol{\alpha} = \mathbf{y}$$

Kernels as similarity rules for scattered readings



Reconstructing a profile with weighted kernel copies



$$\mathbf{K}\alpha = \mathbf{y}, \quad K_{ij} = k(x_i, x_j)$$

weights at $\ell = 0.15$

alternate in sign, up to 4.30 and -3.75

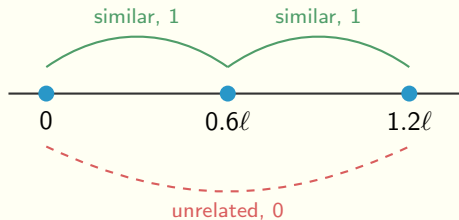
two equal readings one ℓ apart

$1/(1 + \rho) = 0.62$ each

at $x = 0.7$, $\ell = 0.15$

0.216 against the reference 0.393

Positive definiteness of a kernel

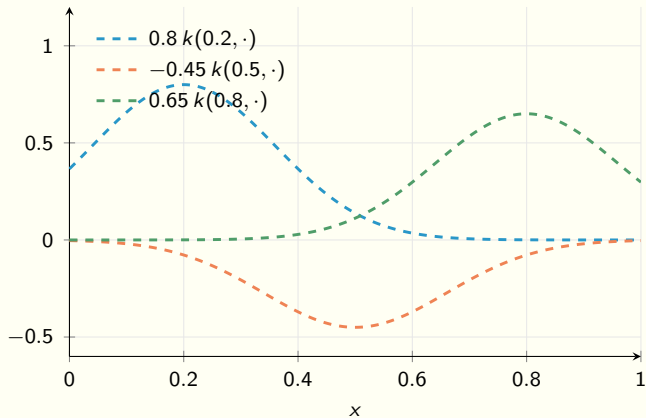


$$\mathbf{K} = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \quad \lambda = 1 - \sqrt{2}, 1, 1 + \sqrt{2}$$

$$\mathbf{w} = (1, -\sqrt{2}, 1), \quad \mathbf{w}^\top \mathbf{K} \mathbf{w} = -1.66$$

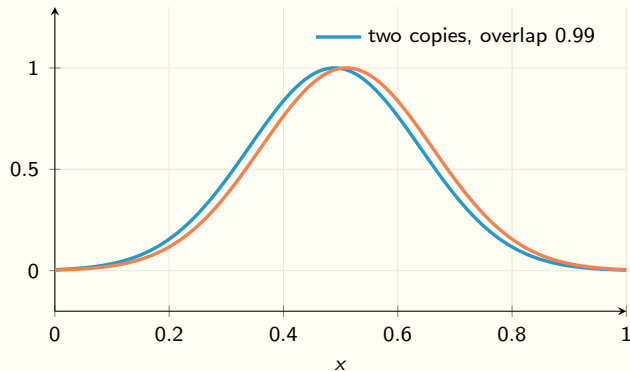
$$k(x, x') = \phi(x)^\top \phi(x') \Rightarrow \mathbf{w}^\top \mathbf{K} \mathbf{w} = \left\| \sum_i w_i \phi(x_i) \right\|^2$$

The reproducing kernel Hilbert space



$$\langle k(x_1, \cdot), k(x_2, \cdot) \rangle_k = k(x_1, x_2)$$

Noise amplification in kernel interpolation



eigenvalues of $\mathbf{K}$, overlap 0.99

1.99 along $(1, 1)$, 0.01 along $(1, -1)$

readings $(1, 1)$

weights near $\frac{1}{2}$ each

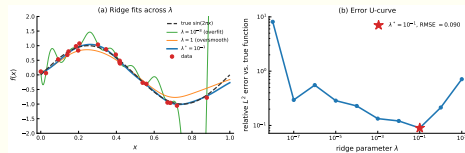
Kernel ridge regression and the choice of λ

$$\min_{f \in \mathcal{H}_k} \sum_{i=1}^n (f(x_i) - y_i)^2 + \lambda \|f\|_k^2$$

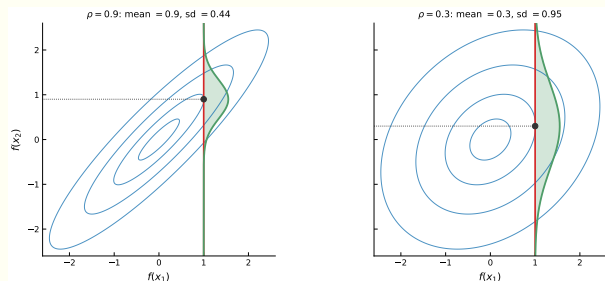
$$(\mathbf{K} + \lambda \mathbf{I})\alpha = \mathbf{y}$$

$\lambda = 0.1$, small divisor
 $0.01 \rightarrow 0.11$

amplification
100 **to** 9.1



Conditioning a Gaussian on an observation



$$f(x_1) = Z_1, \quad f(x_2) = \rho Z_1 + \sqrt{1 - \rho^2} Z_2$$

$$f(x_2) \mid f(x_1) = y \sim \mathcal{N}(\rho y, 1 - \rho^2)$$

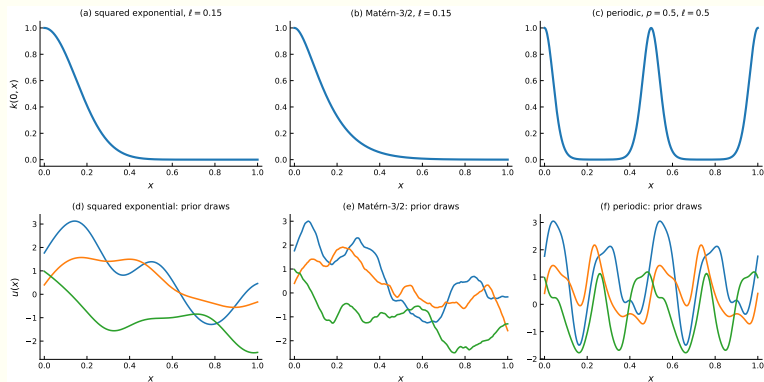
$$\rho = 0.9$$

**mean 0.9y, standard
deviation 0.44**

$$\rho = 0.3$$

**mean 0.3y, standard
deviation 0.95**

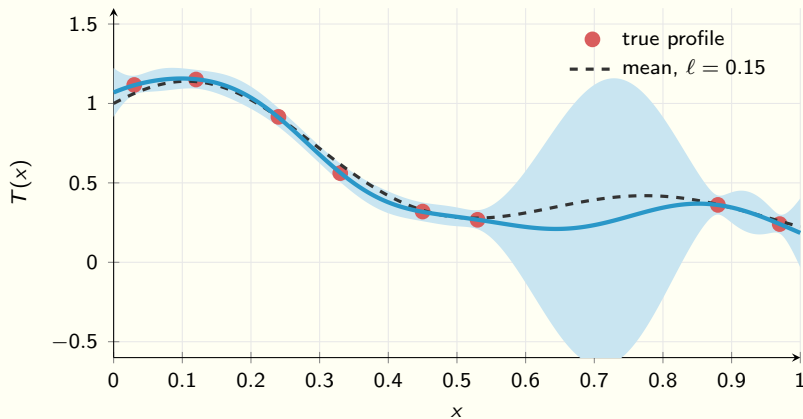
Prior draws from three kernels



mean absolute slope of the
draws
5.2, 8.9, and 20

a kernel method returns
**functions that
resemble its draws,
pulled toward the
data**

Gaussian-process reconstruction of the fin profile



mean

$$\mathbf{k}(x)^\top (\mathbf{K} + \sigma^2 \mathbf{I})^{-1} \mathbf{y}$$

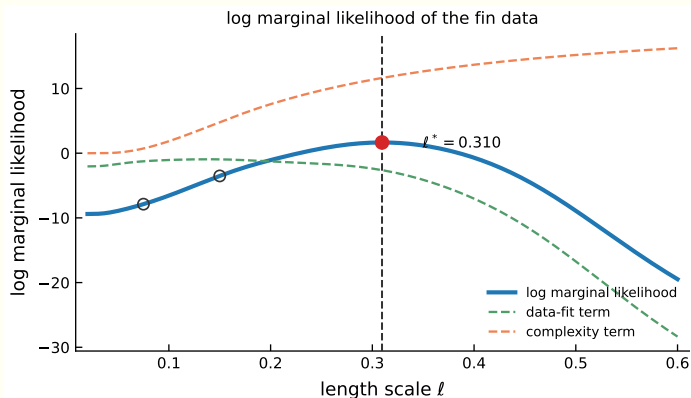
variance

$$k(x, x) - \mathbf{k}(x)^\top (\mathbf{K} + \sigma^2 \mathbf{I})^{-1} \mathbf{k}(x)$$

$\ell = 0.15$

**error 10 percent,
half-width in the gap
0.90**

Choosing the length scale by marginal likelihood



log marginal likelihood at
 $\ell = 0.31$

1.7, the maximum

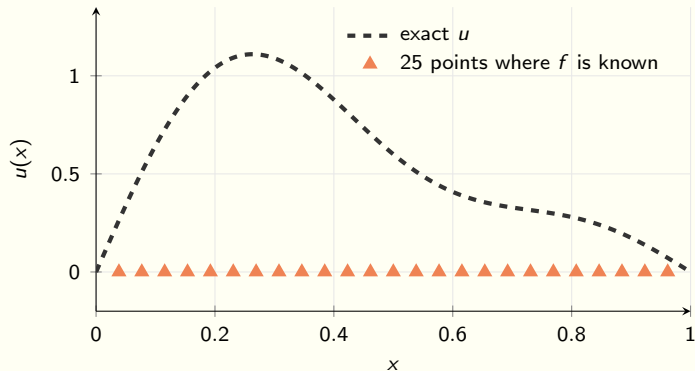
at $\ell = 0.15$ and 0.075

-3.5 and -7.9

with eight readings

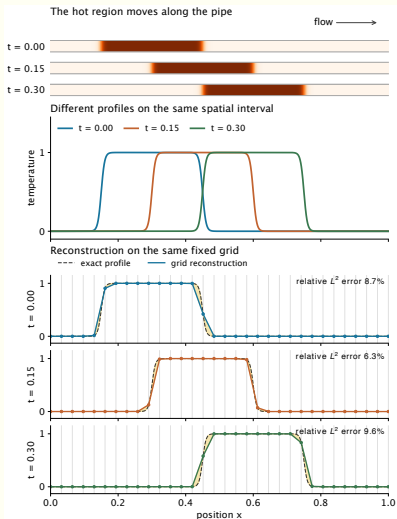
a soft verdict

The kernel prior solves $\mathcal{L}u = f$



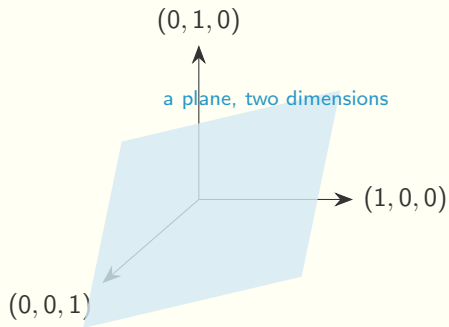
$$\text{cov}(u, f) = \mathcal{L}_{x'} k$$
$$\text{cov}(f, f) = \mathcal{L}_x \mathcal{L}_{x'} k$$

A moving front gives a family of functions



$$T(x, t) = T_0(x - ct), \quad \mathcal{S} = \{T_0(\cdot - ct) \mid 0 \leq t \leq 0.30\}$$

The Kolmogorov n -width



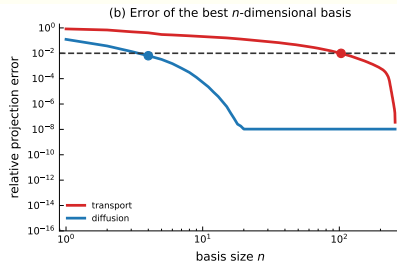
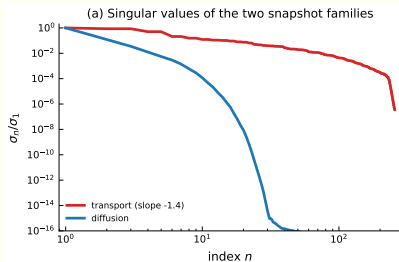
Kolmogorov n -width

$$d_n(\mathcal{M}) = \inf_{\dim V=n} \sup_{f \in \mathcal{M}} \inf_{v \in V} \|f - v\|$$

three unit spikes, squared
distances to any plane
sum to $3 - 2 = 1$

worst spike
at least
 $\sqrt{1/3} = 0.577$

Widths of diffusion and transport families



diffusing pulse, SVD, one percent mean-square snapshot error

four modes

translating pulse, same measure

103 modes

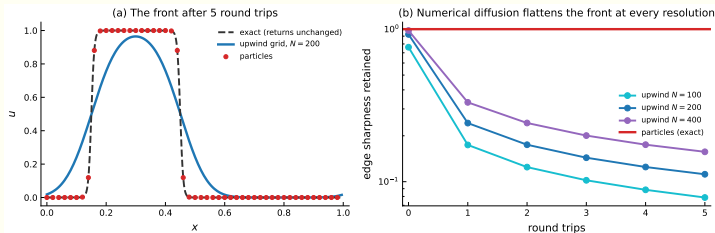
coefficient k under diffusion

shrinks by $e^{-(2\pi k)^2 t}$

coefficient k under translation

changes phase only

Particles carry values with the flow



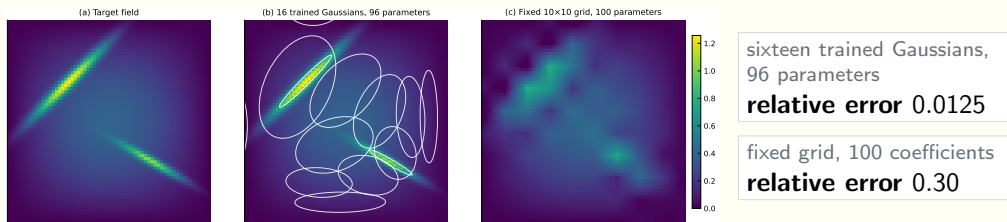
markers after five round trips

the front returns unchanged

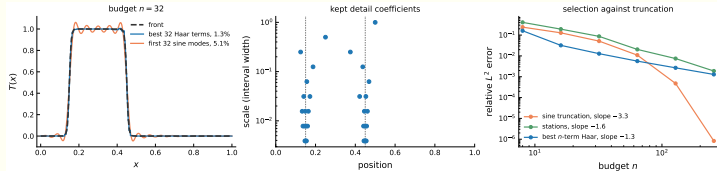
upwind grid at 100, 200, 400 points

keeps 8, 11, and 16 percent of the edge steepness

Gaussian splats with trainable centers and shapes



Selecting coefficients from a fixed dictionary

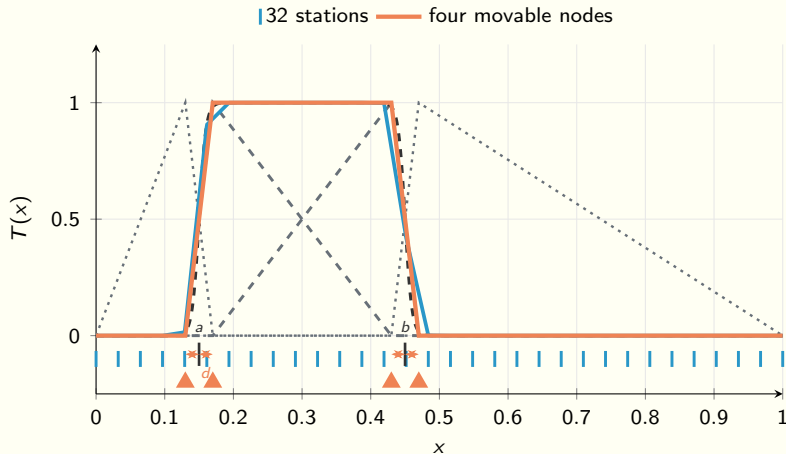


$n = 16$, sines, stations,
largest Haar
12.8, 19.3, 3.2
percent

$n = 32$, largest Haar
against sines
1.3 against 5.1
percent

sines overtake
near $n = 128$

Movable hat nodes on a traveling front



nodes

$a - d, a + d, b - d,$
 $b + d$, with $d = 0.02$

32 stations

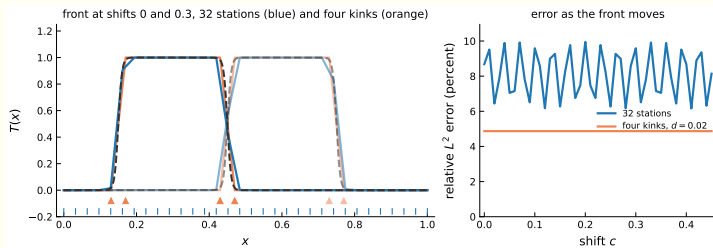
8.7 percent

four nodes

4.9 percent

[Try it live](#)

Fixed nodes and moving nodes



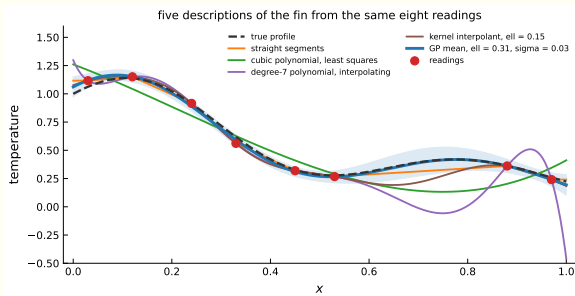
fixed nodes

**a linear solve, with
the projection
theorem's guarantee**

moving nodes

**a non-convex fit,
with no guarantee**

Choosing a representation for the cooling fin



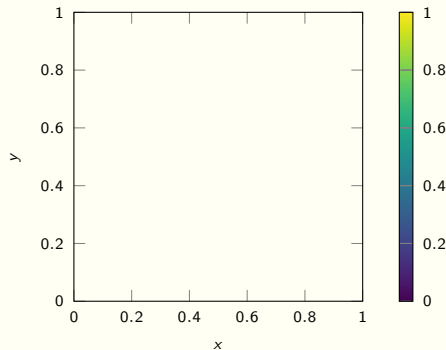
description	error vs truth	leave-one-out
straight segments	6.5 percent	
cubic, least squares	23	
degree-7 interpolation	31	
kernel interpolant, $\ell = 0.15$	11	
Gaussian-process mean, $\ell = 0.31$	3.2	
noise level		

Stored numbers, shape placement, and assumption for each family

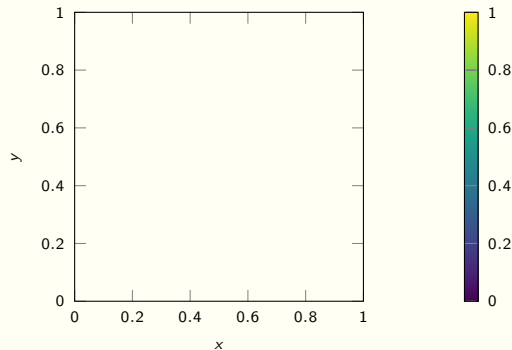
family	stores	shapes placed by	assumes	native regime
stations, grid	values at fixed points	analyst	linear variation between stations	smooth fields, etry
mesh hats	nodal values	analyst and mesher	piecewise-linear variation	complex geom
sine modes	global coefficients	analyst	smooth odd periodic continuation	smooth pinned
B-splines, NURBS	coefficients at fixed knots	analyst and CAD	smoothness across knots	engineered sha
wavelets	selected coefficients and indices	analyst, data selects	few active scales per location	localized featur
POD modes	coefficients on data-derived modes	the snapshot data	a family near a low-dimensional subspace	near-linear fam
sparse dictionary	few atoms and their indices	the optimizer	few active atoms	few active featur
particles, SPH, MPM	positions and payloads	the dynamics	payloads ride their carriers	advection, larg tion
Gaussian splats	centers, covariances, amplitudes	the optimizer	smooth field with concentrated features	concentrated f
level sets	values of an implicit function	analyst and physics	interface as a zero set	moving interfa
kernels, GP	weights, sites, kernel parameters	analyst and data	regularity set by the kernel	scattered data
neural fields	network weights	the optimizer	the network's smoothness bias	later chapters
latent decoders	codes and a decoder	the optimizer	a low-dimensional nonlinear manifold	later chapters

A membrane under a load, $-\Delta u = f$

the load $f(x, y)$



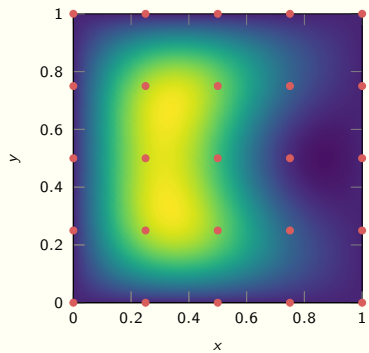
the displacement $u(x, y)$, pinned on the edge



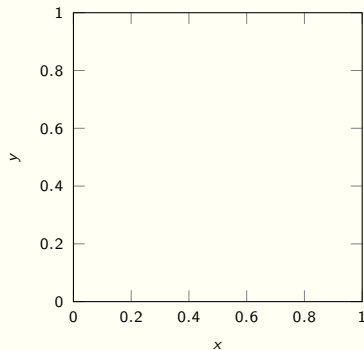
$$-\Delta u = -\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) = f \text{ on the unit square,} \quad u = 0 \text{ on its edge}$$

Readings on a grid and bilinear reconstruction

the field u and the 5×5 stations

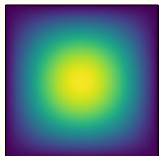


bilinear reconstruction from 25 readings

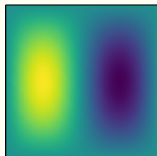


Sine modes and hats in two dimensions

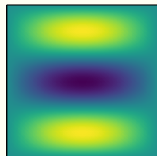
$\sin(\pi x) \sin(\pi y)$



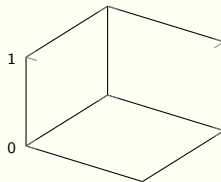
$\sin(2\pi x) \sin(\pi y)$



$\sin(\pi x) \sin(3\pi y)$



a hat on a triangle mesh



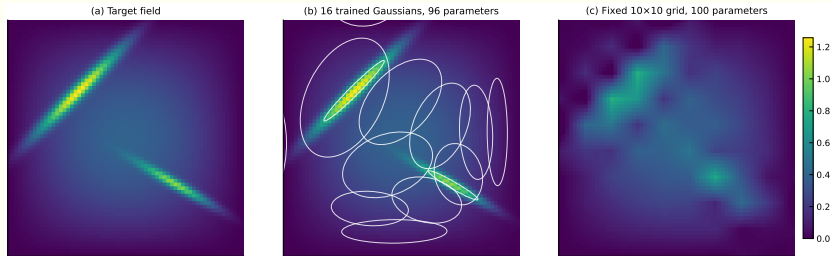
$$u(x, y) = \sum_{j,k} a_{jk} \sin(j\pi x) \sin(k\pi y),$$

$$u_h(x, y) = \sum_i u(\mathbf{x}_i) \phi_i(x, y)$$

the membrane's weights

$$a_{11} = 0.8, a_{21} = 0.4, \\ a_{13} = 0.2$$

What changes in two and three dimensions



17 stations per direction
**17 on a line, 289 on a
plate, 4,913 in a solid**

Summary

- 1 Pointwise sampling
- 2 Functions as vectors
- 3 Norms and the meaning of error
- 4 Function spaces, L^2 , H^1 , and Hilbert spaces
- 5 Inner products, projection, and orthogonality
- 6 Changing the basis
- 7 Convergence, density, and completeness
- 8 Kernels and point evaluation
- 9 Noisy observations, ridge regression, and Gaussian conditioning
- 10 Approximating a family of functions
- 11 Adaptive representations
- 12 Choosing a representation for the cooling fin
- 13 From functions to fields